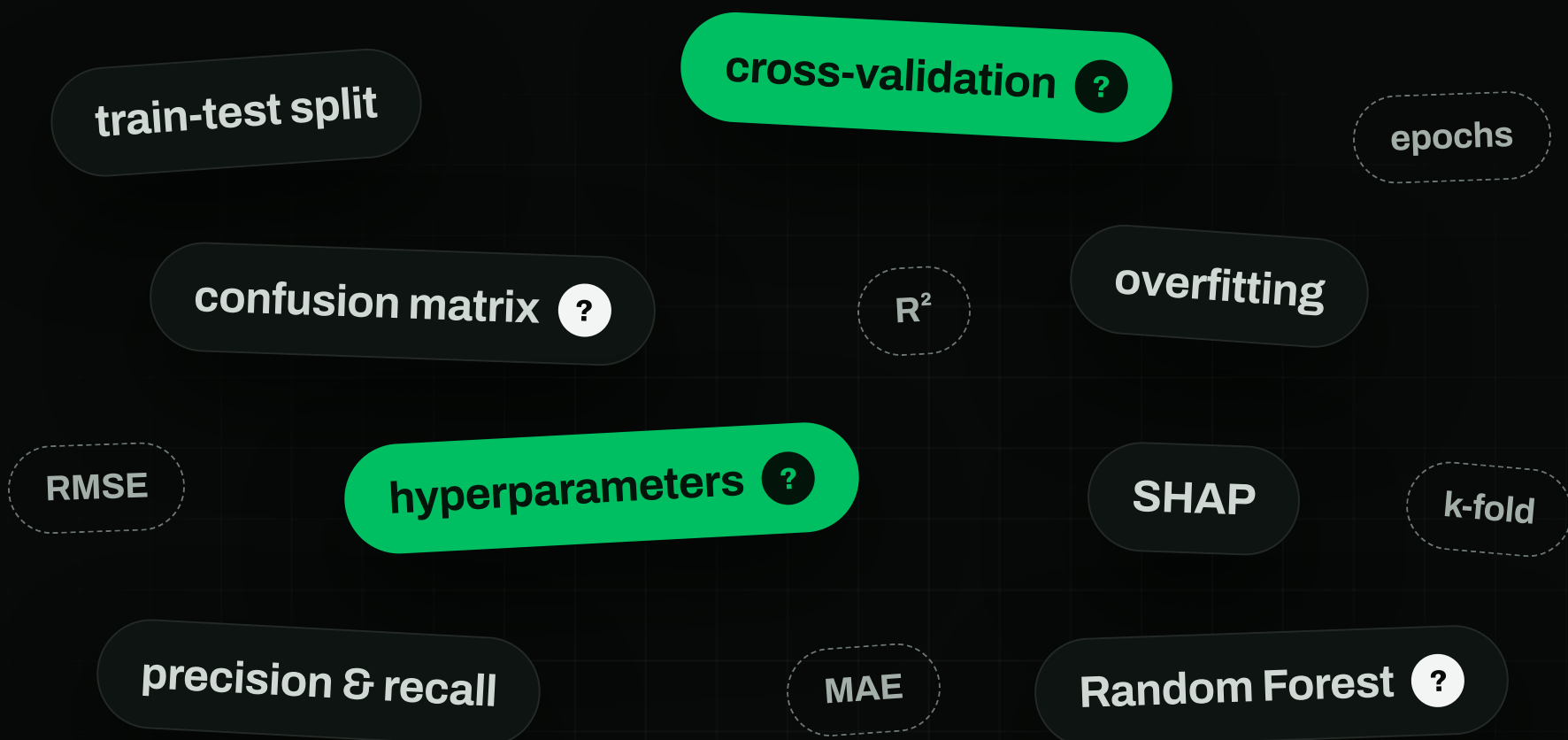


Still confused by the **jargon** when people talk about Machine Learning?



You open an ML-based research paper... and get stuck on the **terms**.

RESEARCH ARTICLE

CIVIL ENGINEERING · ML

Predicting the compressive strength of concrete using ensemble machine learning

A. Author, B. Author · Department of Civil Engineering

ABSTRACT The dataset was divided using an 80/20 **train-test split**, and **hyperparameters** were tuned with 10-fold **cross-validation**. The **Random Forest** model achieved an **R²** of 0.91 with an **RMSE** of 4.8 MPa, showing no sign of **overfitting**. **SHAP** analysis was used to interpret the influence of each mix parameter...

?

?



Or perhaps you understand the **engineering problem perfectly**, but the ML terminology makes the paper feel **unnecessarily difficult**.

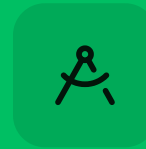
This course is designed **for you.**

As a Civil Engineer, you don't need to become a computer scientist to understand and start using Machine Learning.



YOU DON'T NEED TO

~~Become a computer
scientist~~



YOU NEED TO

**Understand how ML
fits into an
engineering workflow**

Step by step, from **problem** to **decision**.



01 Engineering Problem

Which mix inputs drive concrete strength?



02 Data

Explore, clean and question the dataset.



03 Model

Let the model learn the pattern from examples.



04 Validation

Test it honestly, on data it has never seen.



05 Interpretation

Does what it learned make engineering sense?



06 Engineering Decision

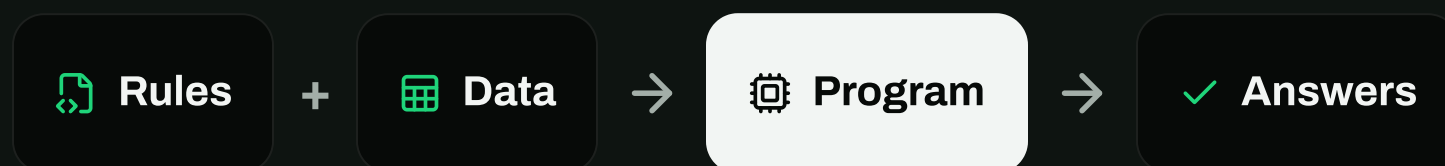
Act on the result, with its uncertainty in view.

What Machine Learning **actually** means.

And how it differs from conventional programming.

Conventional programming

You write the rules.



Machine Learning

The model learns the rules from examples.



ALSO ON DAY ONE

Working with engineering datasets using Python

Load, explore and plot real data in Google Colab. No installs.

#	Cement	Water	Age (d)	Strength (MPa)
0	540.0	162.0	28	79.99
1	332.5	228.0	270	40.27
2	198.6	192.0	360	44.30
3	266.0	228.0	90	47.03

How a model is trained and tested.

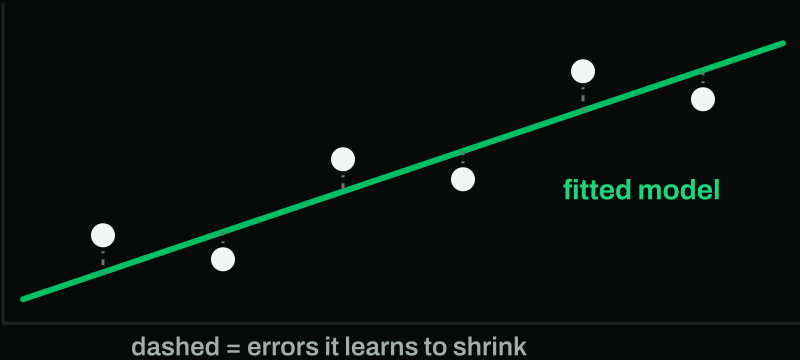
1 What train-test splitting really does

Hold part of the data back, so the model is judged on examples it has never seen.



2 What model training means

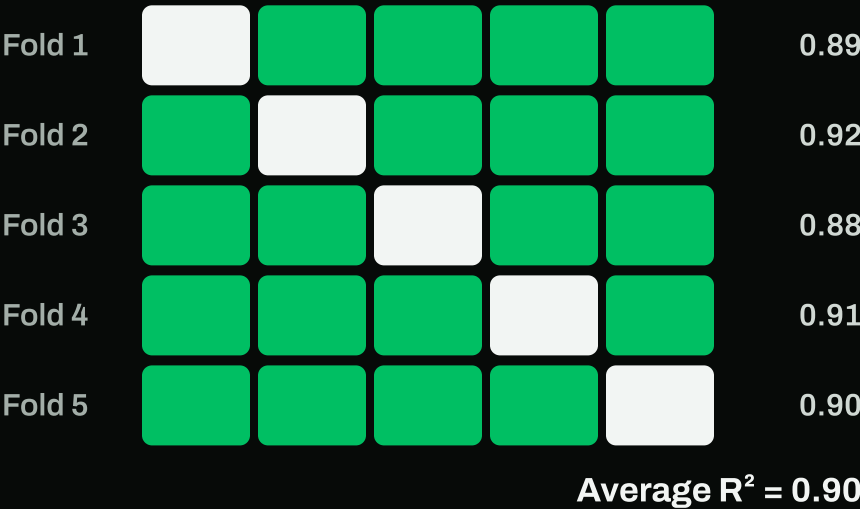
The model adjusts itself, step by step, to reduce its error on the training data.



3 Why we use k-fold cross-validation

Rotate which part is held out, so every data point gets tested once. The average score is far more trustworthy than a single split.

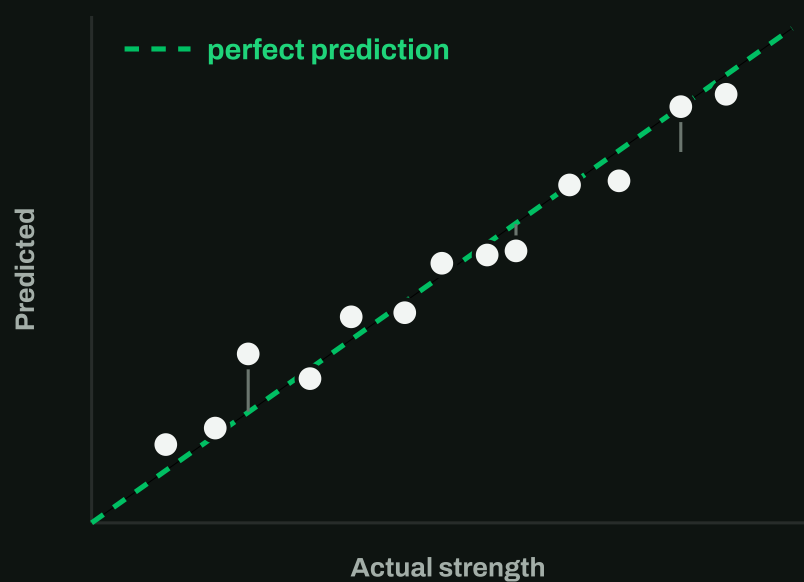
■ Train ■ Test fold



Read a model's scorecard.

What the numbers in every results table actually tell you.

Regression: MAE, RMSE, R²



- MAE** · the average size of an error
- RMSE** · like MAE, but big misses count more
- R²** · how much of the variation is explained

Classification: the confusion matrix

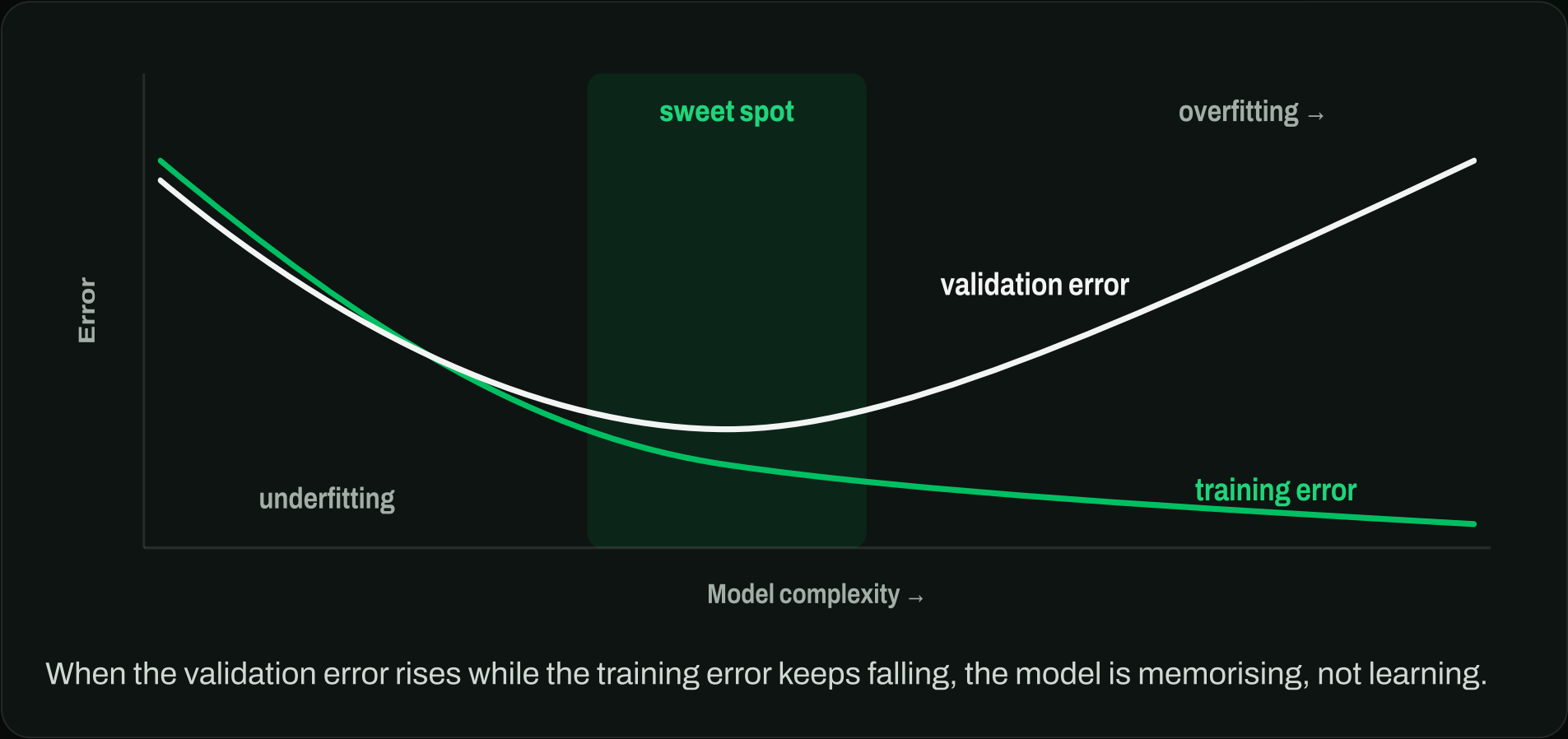
	Predicted unsafe	Predicted safe
Actually unsafe	42 caught	5 missed
Actually safe	3 false alarm	50 correct

- Precision** · of the slopes flagged unsafe, how many really were? 42 / 45
- Recall** · of the truly unsafe slopes, how many did we catch? 42 / 47



Why it matters: these numbers tell you whether a paper's model is genuinely good, or just looks good.

Why models **overfit**, and how to detect it.



THE MODELS YOU'LL BUILD, AT A PRACTICAL LEVEL

Decision Trees

A chain of yes/no questions about the inputs.

Random Forests

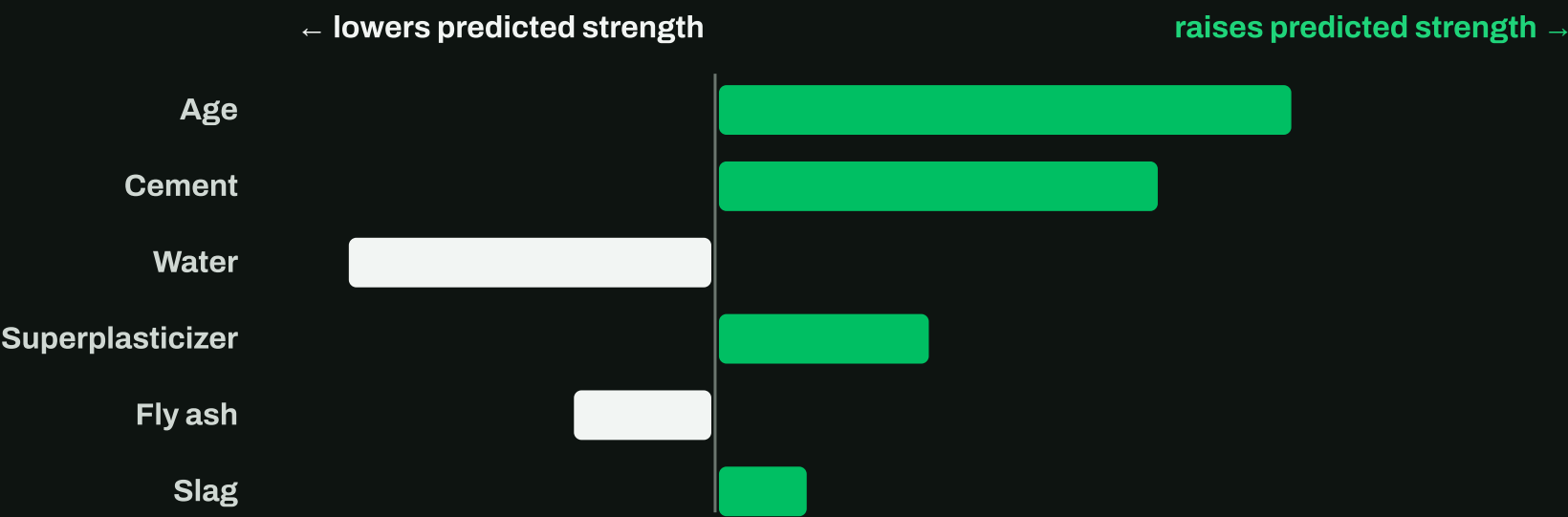
Many trees vote, so the answer is steadier.

Neural Networks

Layers of simple units that learn complex patterns.

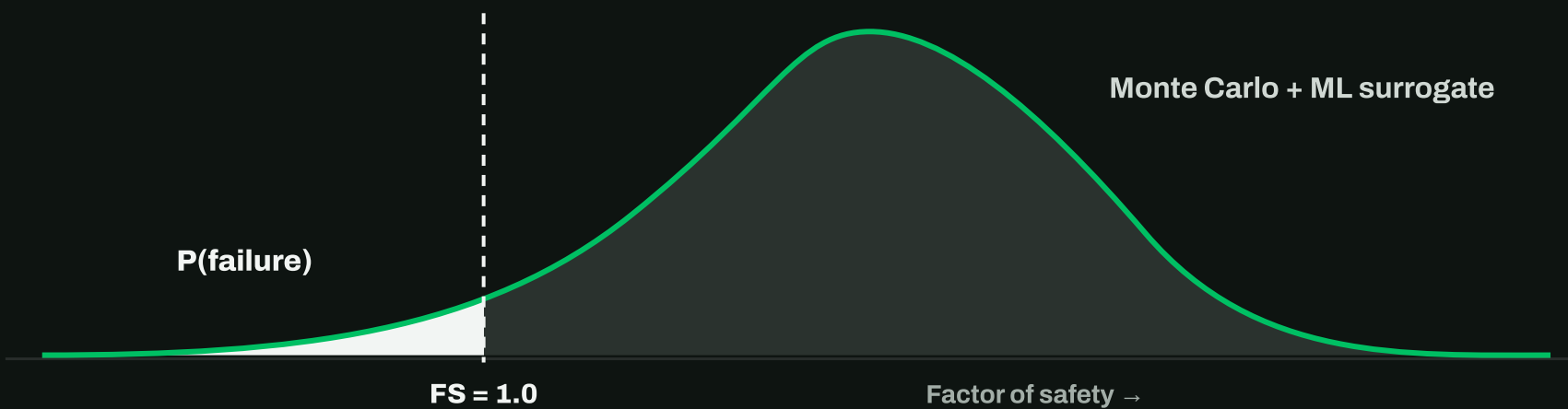
Explain the model. Quantify the risk.

SHAP & Explainable AI



See which inputs pushed a prediction up or down, and check that it makes engineering sense (more water, lower strength).

Uncertainty & reliability analysis



Go beyond a single number: connect ML to uncertainty and estimate the probability of failure.

Learned on **real Civil Engineering problems** and datasets.



Materials

e.g. concrete compressive strength



Buildings

e.g. building energy performance



Manufacturing

e.g. production and quality data



Water & Environmental

e.g. water quality classification



Geotechnics

e.g. slope stability and soil response



All in Google Colab

Python notebooks in your browser. Nothing to install.

Next time you open an ML paper, it won't feel like a foreign language.

THE JARGON	WHAT IT ACTUALLY MEANS
train-test split	→ Data held back for an honest test
cross-validation	→ Testing several times on different held-out parts
overfitting	→ Memorising the training data instead of the pattern
confusion matrix	→ A table of right and wrong classifications
hyperparameters	→ Settings you choose before training starts

- 01

Get acquainted with Machine Learning
- 02

Understand the jargon
- 03

Learn the workflow
- 04

Start applying it to your own problems

Let's make Machine Learning part of your **Civil Engineering** toolkit.

 No prior Python or Machine Learning experience required



YOUR INSTRUCTOR

Dr. Abdul Waris

Research Associate, Universität Siegen · Ph.D., IIT Hyderabad

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